

EM 505 - Managing Engineering and Technology (Online)

EMU COLLEGE OF TECHNOLOGY - School of Engineering Technology

Prerequisites: none

Credit hours: 3 Semester Hours

Professor: Richard H Townsend rtownse2@emich.edu

Office Hours: by appointment

Communication:

E-mails are the preferred communication channel for asking questions or for scheduling a phone or Skype meeting

Course Description

An investigation of the functions and problems unique to managing engineering and technical operations. Topics include: engineering ethics and professionalism; managing research, design and other technical operations; technology impact; and the role of the technical professional.

Required Texts and Assessments

The required text books are:

1. Morse, L., & Babcock, D. (2010). Managing engineering and technology. (5th ed.). Upper Saddle River, NJ: Prentice Hall. ISBN-10: 0136098096.

2. Newton, L., Englehardt, E., & Pritchard, M. (2010). Taking sides: clashing views in business ethics and society. (11th ed.). New York, NY: McGraw Hill. ISBN-10: 0073527319

Course Objectives

- (a) Develop an understanding of the nature and role of technology.
- (b) Develop an understanding of the nature and role of engineering management.
- (c) Explain the functions of technology management.
- (d) Explain the different technology management roles throughout the product life cycle

Grading Weighting

Unit exams - 17%

Journal synopsis - 20%

Objective questions - 3%

Team and class discussion - 15%

Final exam - 15%

Threaded discussion - 15%

Skype, Intro discussions, Preparation quiz, Final discussion, Peer evaluation - 15

Course Outline and Schedule

Unit Schedule: Click Due Dates tab within Course Home

Unit 1.

Ch. 1: Nature and Roles of Engineering and Management

Ch. 2: Historical Development of Engineering Management

Unit 1 Threaded Discussion

Unit 1 Exam

Unit 1 Journal

Unit 1 Objective Questions

Unit 1 Team Discussion

Unit 2.

Ch. 3: Planning and Forecasting

Ch. 4: Decision Making

Ch. 5: Organizing

Unit 2 Threaded Discussion

Unit 2 Exam

Unit 2 Journal

Unit 2 Objective Questions

Unit 2 Team Discussion

Unit 2 Management Philosophy Statement

Unit 3.

Ch. 6: Human Aspects of Organizing

Ch. 7: Leading Technical People

Unit 3 Threaded Discussion

Unit 3 Exam

Unit 3 Journal

Unit 3 Team Discussion

Unit 3 Management Philosophy Statement

Unit 4.

Ch. 8: Controlling

Ch. 9: Managing Research and Development

Ch. 10: Managing Engineering Design

Unit 4 Threaded Discussion

Unit 4 Exam

Unit 4 Journal

Unit 4 Management Philosophy Statement

Unit 5.

Ch. 11: Planning Production Activity

Ch. 12: Managing Production Operations

Ch. 13: Engineers in Marketing and Service Activities

Unit 5 Threaded Discussion

Unit 5 Exam

Unit 5 Journal

Unit 5 Team Discussion

Unit 5 Management Philosophy Statement

Unit 6.

Ch. 14: Project Planning and Acquisition.

Ch. 15: Project Organization, Leadership, and Control

Ch. 16: Engineering Ethics

Unit 6 Threaded Discussion

Unit 6 Exam

Unit 6 Journal

Unit 7.

Ch. 17: Achieving Effectiveness as an Engineer

Ch. 18: Globalization and Challenges for the Future

Unit 7 Threaded Discussion

Unit 7 Exam

Unit 7 Journal

Unit 7 Management Philosophy Statement

Unit 8.

EM505 Final Discussion

Final Exam

General Policies

A. Problems.

Students should discuss problems or misunderstandings about their course work directly with the instructor via e-mail or telephone and not via broadcast e-mails to the entire class.

B. Online Participation and Conduct.

Interaction via threaded discussions is an active learning component of the class. Plan to contribute to every discussion topic with a "quality" contribution -- i.e., clear, constructive, introspective and relevant input. Certainly I am expecting participation that is beyond statements such as: "I agree."

Our instructors and the University can not and will not be responsible for the content of any personal messages that are sent from one student to other students via email or threaded discussion. To maintain privacy, it is best NOT to share phone numbers or addresses to classmates unless necessary for team communications. It is expected that all students will adhere to accepted codes of ethical, personal, and civil conduct when conversing online, using e-mail, or engaging in any real time online chat sessions. Failure to abide by such codes of conduct and etiquette may result in withdrawal from the course and a failing grade.

When you send emails related to a class, you should put informational codes in the subject line of your email messages. For example, include the: initials of your full name, course number, and the subject of the message (such as the name of an assignment that you may be asking about); such as, Subject: BWB505Homework1. Also, do not forget to include your full name in the text of your email message.

C. Grammar, Spelling, Writing Style, and Format.

Good grammar, spelling, writing style, and APA format is required for formal papers, per School of Engineering Technology (SET) policy. Students should follow APA (American Psychological Association) technical writing style and format guidelines for papers submitted in all SET classes. Some professors will emphasize use of APA guidelines more than others. It is recommended that you get a copy of the 6th edition of the APA Publication Manual as soon as possible, as you will use it throughout your degree. Unless otherwise specified by your professor, you should write your papers in final manuscript format (see Chapter 6, APA manual), NOT copy manuscript format (which is used when submitting articles to journal editors for pre-publication review and editing. You should especially be familiar with traditional and electronic referencing guidelines for quoting, paraphrasing, and referencing information that you use, to help avoid Academic Dishonesty problems, as mentioned above. See <http://www.apa.org/journals/acorner.html#pubmanual> for more specific information and examples.

You should consider all of your communications with the class and professor to be professional. Good grammar and spelling is important for a professional in the business world. If that is difficult for you, we suggest that you compose threaded discussion responses or email messages in a word processing

program where you can check spelling and grammar, and revise your writing. Then you can cut and paste the work into the threaded discussion or email message.

D. Academic Dishonesty.

The policy of the School of Engineering Technology is that any student found to have engaged in any activity constituting academic dishonesty will receive an "F" for the course in which the activity occurred. This policy relates to all forms of work associated with the course requirements (tests, lab work, papers and other assignments). Academic dishonesty includes: cheating on quizzes and exams, copying, or taking credit for another person's work, plagiarism, etc. Graduate students enrolled at EMU should be aware of academic dishonesty policies and consequences, and examples of what constitutes academic dishonesty, from EMU Academic Affairs, EMU Student Judicial Services, EMU Graduate School, and the School of Engineering Technology.

Our managerial approach regarding minor occurrences of academic dishonesty in this course is to inform the involved student(s) about the problem, give a warning to the student(s) of the possible consequences, and share an example of the problem with the rest of the class, as a learning experience for the whole class. The consequences may be more severe (such as failure of the assignment or the entire course), and without a warning, for major occurrences and/or second minor occurrences of academic dishonesty that occur in this course, or that are on record for other courses that you may have taken. Case records of academic dishonesty are kept in the SET department and/or the Student Judicial Services office.

I will be using TurnItIn.com to check papers for originality. Do not plagiarize. Do cite references when quoting from other sources.

E. Incomplete Final Grades (I): Incomplete grades will be given only in accordance with EMU's guidelines, which are stated in the 2005-2007 Graduate Catalog (p.19), at <http://www.emich.edu/public/catalogs/2005-2007/grad/20051gfront.pdf>, as follows: I (Incomplete) Grades - An "I" grade may be awarded when illness or other unavoidable extenuating circumstances prevent completion of a course, provided that academic performance for the completed portion of the course (50% or more) merits a grade of B or better.

F. Student and Exchange Visitor Information Service (SEVIS).

For students who hold an F or J visa, to avoid being out-of-status, it is your responsibility to report within 10 days to the Office of International Students (OIS), 244 Student Center, any change in name, current address of residence, academic status (full or part-time enrollment), date of program completion, program/major change, level change (undergraduate to graduate), change in funding source (employment or graduate assistant position), probation or disciplinary action leading to suspension due

to conviction of a crime. Effective January 1, 2003, EMU has 21 days from the date of the reported event to inform the U. S. Government -- Department of Homeland Security, Bureau of Border and Transportation Security and the Bureau of Citizenship and Immigration Services -- SEVIS database (functions of the formerly known as Immigration and Naturalization Service are a part of two new offices effective March 1, 2003). OIS enters the data into a real-time database and new documents or I-20s are then issued to you. Remember that dropping or withdrawing from course(s) may put you below full-time enrollment. Before a drop/withdrawal change is made that reduces enrollment below full-time status, you must receive permission from the OIS. If you plan to transfer to another university, this process must begin before the end of your last semester at EMU, contact the Office of International Students. Noncompliance could result in actions leading up to a fine, arrest, or deportation per the U.S. Government.

Per SEVIS regulations, International students who hold an F or J visa can only take one fully online class (no face-to-face meetings) a semester. Hybrid classes have face-to-face meetings, so are allowed.

G. Observance of Religious Holidays.

Eastern Michigan University recognizes the right of students to observe religious holidays without penalty to the student. We have students from a wide variety of cultures and countries in our EM program. We will follow a functional policy for this class that can accommodate everyone's desire to observe a religious holiday. Students in EM 505 do not have to do class work on their religious holiday, but deadline dates for assignments will not be extended. If a student's religious holiday occurs on the last day of a unit, the student should not wait until the last day of the unit to complete their assignments. Per course policy, late assignments will not be accepted. Students are encouraged to not wait until the last minute to submit assignments.

H. Personal Involvement and Time.

Students often have a wrong perception that taking a course online is easier since they don't have to physically go to a classroom. When taking an online course there are some things that become easier for students. For example, it becomes easier NOT to go to class. It also becomes easier to get behind. This medium gives students flexibility in attending class--but remember, while taking a course online may be more convenient, it is not necessarily easier!

Be aware of the general guidelines to professors for student work loads: 45 hours of work per credit hour per semester. This is a 3 credit class, so we expect, on average, that students will spend 135 hours of total work time in this course. Students with more graduate-level and online learning experience tend to be more efficient and require less time--some students will require a great deal more time to get their school work completed. Be prepared for the work involved, as it may require 10-15 (or more) hours of your time each week.

This is an Engineering Management course, part of the EM program, taken by EM masters and doctoral students. Engineering Managers are responsible for planning, organizing, staffing, leading, and controlling engineering projects and personnel. They must lead by example, and make sure that objectives are met and completed on time. Poor planning and procrastination, such as trying to complete all work at the last minute, leads to poor quality and missed deadlines because there is no buffer available to deal with unexpected problems. Many of our EM students lead very busy professional lives and have significant work and family responsibilities in addition to going to school--yet they do their school work well, and get it done on time (often early). If a student in our program can not organize themselves well enough to get their school work done on time, then they will certainly have a problem organizing and leading other engineers and engineering projects, and probably should not be in this program. Therefore, we will give little leeway when it comes to assignments that are not submitted on time--and will suggest to chronically late students that they seek a different degree and profession.

Specific Course Policies, Guidelines, and Grading

A. Course Unit Timing

This course consists of 8 units, which are scheduled from Mondays through Sundays. Most units will be scheduled across two full weeks with assignments due by the end of each unit on Sunday nights, by 11:59 PM ET. Exceptions to assignments being due by the end of the Unit include team debate introductory statements and team member peer evaluations. No assignments required to be completed by the end of a Unit will be accepted after the close of that Unit. All completed units will remain open for review until the end of the semester.

B. Access to Course

Our course consists of 8 units, which are scheduled from Mondays through Sundays. Most units will be scheduled across two full weeks with the exception of Unit 1, which starts on a Wednesday, and Unit 8, which ends on a Friday. All assignments will be due at the end of each unit on Sunday nights, by 11:59 PM ET. No assignments required to be completed for a unit will be accepted after the close of that unit. All completed units will remain open for review, until the end of the semester.

C. Email to Instructor

E-mail is the preferred communication method for this class. Please send all email directly to Prof. Townsend at his personal email account, at rtownsend3@my.emich.edu. Include your full name, course number, and the subject of the message in the e-mail subject line. For example Dr. Booker's name is Bryan W. Booker so his subject line might read: BWB547Presentation1.doc. I plan to check my E-mail

frequently throughout each week. If you procrastinate then you may not receive the timely feedback that you need for completing your work by the due date.

D. Course Format and Activities.

The general format of this class will be through online guidance from the professor and students' submission of assignments, which include: reading presentation assignments and quizzes, case study discussions, peer evaluations of fellow students' work and taking a final exam.

E. Assignment Software, Submission to Drop Box, and File Names.

There are so many different word processing programs that reading assignments can become a problem in online courses if there is no standard policy set for them. It is preferred that you submit papers in MS Word file format. Formats that are compatible with MS Word include earlier Word versions, Microsoft Works, Word Perfect 5 or 6, and Rich Text Format. If necessary, you can compose the work in your non-compatible word processor and simply cut and paste it into an email message--although this may cause a problem if formatting is important. If using MS Project software, please submit your work in MS Project 2007 or 2010 format. You may also send single or multiple files in .zip compressed format.

Submit assignments to the Unit Drop Box as directed for each Unit. The file names used for assignments should follow a coding format similar to that used for course-related emails. For example, include the: initials of your full name, course number, and the name of the assignment with its file type; therefore, the file name of an assignment would look like: BWB505Presentation1.doc.

I may use TurnItIn.com to check papers for originality after they are submitted to the drop box. Details about www.turnitin.com will be

provided when needed.

F. Assignment Delivery and Technical Problems.

Since you will have more than two weeks and three weekends available for completing your work throughout most of the course, we expect you to get it done on time. No assignments will be accepted after the close of a unit. Written paper assignments are to be submitted using the Unit Assignment Drop Box, and are due one week prior to the close of a unit (see schedule in syllabus). Written paper assignments will not be accepted after the close of a unit. Other work such as threaded discussions, quizzes, and exams are submitted using those course tools, which automatically close at the end of a unit. The final exam (and any other assignments due in our Unit 8 final week) will not be accepted late. The units are planned to be available to you at least two days prior to the beginning of a unit. It is recommended that you plan on getting your work done during the first half of the unit, and plan on

encountering unexpected problems. You have been made aware of the consequences of poor planning, procrastination, and not getting your work done on time.

We encourage you to make hard copies of your assignments and communications with your instructors. Computer crashes and disasters can happen--we expect that you practice good risk management behavior, just like good engineering managers. We do not view ISP problems as an excuse for turning work in late. Alternative methods of submitting assignments include submitting your work via facsimile (Fax) to Dr. Booker at 734.487.8755 (EMU) or e-mailing me with an alternate proposal for submitting your work.

Note: It is your responsibility to access the course on another computer in the event of hardware or software problems. Our guidance from the university is to not accept excuses for missing assignments due to technical difficulties.

G. Quizzes and Exams

Most of the quizzes and exams have a maximum duration to allow two minutes per question. The typical standard for professional-level exams, such as certification or licensure exams, is about 60 to 90 seconds per question. The quizzes and exams in this course are focused on specific information from sections of specific books--whereas certification and licensure exams cover a wider body of professional knowledge, covered by numerous books and other material.

Like some professional exams, your quizzes and exams in this class are open-book like those professional exams. It is recommended that you prepare in advance of completing your quizzes and exams - study the material, annotate your text while reading, write notes, and apply the test-preparation studying techniques that work best for you.

H. Evaluation and Grading.

Course grades will be determined by the percentage of total points attained for various work assignments completed throughout the semester, based upon the following scale:

100 to 93% = A

92.99 to 90% = A-

89.99 to 87% = B+

86.99 to 83% = B

82.99 to 80% = B-

79.99 to 77% = C+

76.99 to 73% = C

72.99 to 70% = C-

(no D grades, based on Grad School requirements)

69.99 and below = F

Final course grades will be based primarily (but not necessarily only) on the quantitative scale shown above--but are subject to adjustment based on the judgment of your professor in consideration of qualitative factors, including academic dishonesty and displays of other unethical, unprofessional, or uncivil behaviors that are unbecoming of a good engineering manager and graduate student. (You must to be prepared for the reality of professional life in business and industry--there is usually more to getting a good performance evaluation from your boss and peers than just production of work, other factors include: (a) personal professional skills, (b) professional communication skills, and (c) ethics and emotional intelligence.) (a) (b) and (c) above are of vital importance to becoming successful as an Engineering Manager, and why they are three of our EM Program Student Learning Objectives. We expect to see them demonstrated by students in our classes.

I. Debate-Cases

In some classes, during your Engineering Management program, you will participate in Case Studies solutions and debating various issues. Read the Unit Debate-Case to learn how to analyze case study and participate in debate.

J. EMU Writing Support

The University Writing Center (115 Halle Library; 487-0694) offers one-to-one writing consulting for both undergraduate and graduate students. Students can make appointments or drop in between the hours of 10 a.m. and 6 p.m. Mondays through Thursdays and from 11 a.m. to 4 p.m. on Fridays. Students should bring a draft of what they're working on and their assignment sheet. The UWC opens for the Winter 2013 semester on Monday, January 14 and will close on Friday, April 19.

The UWC also offers small group workshops on various topics related to writing (e.g., Organizing Your Writing; Incorporating Evidence; Revising Your Writing; Conquering Commas; Finding and Fixing Errors). Workshops are offered at different times in the UWC. Visit the UWC page

(<http://www.emich.edu/english/writing-center>) to see our workshop calendar. To register for a workshop, click the link from the UWC page for the type of workshop you wish to attend.

The UWC also has several satellite sites across campus. These satellites provide writing support to students within the various colleges. For more information about our satellite locations and hours, visit the UWC web site: <http://www.emich.edu/english/writing-center>.

The Academic Projects Center (116 Halle Library) also offers one-to-one writing consulting for students, in addition to consulting on research and technology-related issues. The APC is open 11 a.m. to 5 p.m. Mondays through Thursdays for drop-in consultations. Additional information about the APC can be found at <http://www.emich.edu/apc>. Students visiting the Academic Projects Center or any of the satellites of the University Writing Center should also bring with them a draft of what they're working on and their assignment sheet.